Lec 5
String, indexing, slicing

What are strings?

String is a collection of unicode characters.
unicode characters :- capital alph. (A-Z)
small alph. (a-z)
Digits - (0-9)
Special characters - (@, $, %

- String is written inside single, double or tripple quotes (inversed comma)

eg. 'hello world' - single quotes
"hello world" - Double quotes
"""hello world""" - tripple quotes

```
name = "ganesh"      → string
age = 25             → int
weight = 65.9        → float
Eligible_for_vote = True  → Boolean
num1 = 2+3j          → complex
```

- To access the entire string or display the entire string we use that string variable to display it.

Date ___ / ___ / ____

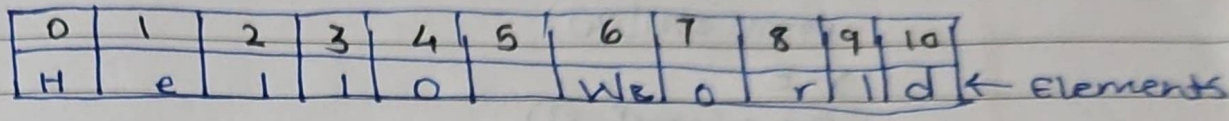
Eg. str1 = "hello world"
print(str1)

◇ **Indexing :**

Using indexing, we can access / display the individual elements of the string.

- string is a sequence of characters

Eg.

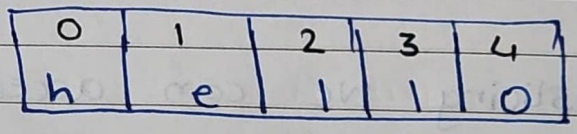


- Every element has position, called as indexing.

Two types of indexing :

1) **Positive Indexing :** It starts from 0.

Eg:-




```
str1 = "hello"
print(str1[0])
```

Syntax : print(variable_name[position])

```
print(str1[0])
> h
print(str1[1])
> e
print(str1[2])
> l
print(str1[3])
> l
print(str1[4])
> o
```

Date __/__/____

2] Negative Indexing :

string starts from -1

Eg.

-5	-4	-3	-2	-1
H	e	l	l	o

```
str1 = "Hello"
print(str1[-5])
> H
print(str1[-4])
> e
Print(str1[-3])
> l
Print(str1[-2])
> l
Print(str1[-1])
> o
```

Slicing :

Using slicing we can access substring (part of string)

- we can print group of element

Slicing syntax :

String-name [start : stop : step]

1. **start** : Start index for slicing
2. **stop** : stop index, slicing stops here, stop index
3. **step** : Interval between characters.

Eg. str1 = "Hello world."

0	1	2	3	4	5	6	7	8	9
H	e	l	l	o	W	o	r	l	d

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```
print(strg[0 : 5 :1])
> Hello
print (strg[0 :12 :1])  → NO error will been shown
> Hello world                as no range.
print (strg[ :  : ])
> Hello world
print (strg[0 :5 :2])
> Hlo
print(strg[5: 10 :2])
> wrd
```